$\frac{d}{dx}(\sec x) = \sec x \tan x$	$\int \sec x \tan x \, dx = \sec x + c$
$\frac{d}{dx}(-\text{cosec} x) = \text{cosec} x \cot x$	$\int \text{cosec} x \cot x \, dx = -\text{cosec} x + c$
$\frac{d}{dx}(e^x) = e^x$	$\int e^x dx = e^x + c$
$\frac{d}{dx}\left(\frac{a^x}{\log a}\right) = a^x$	$\int a^x dx = \frac{a^x}{\log a} + c$
$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$	$\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + c$
$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$	$\int \frac{1}{1+x^2} dx = \tan^{-1} x + c$

எடுத்துக்காட்டு 11.1

தொகையிடுக :

(i) x^{10} (ii) $\frac{1}{x^{10}}$ (iii) $\sqrt{x}$ (iv) $\frac{1}{\sqrt{x}}$

தீர்வு

(i) $\int x^n dx = \frac{x^{n+1}}{n+1} + c, \quad n \neq -1.$

$n = 10$ எனப்பிரதியிட,

$$\int x^{10} dx = \frac{x^{10+1}}{10+1} + c = \frac{x^{11}}{11} + c$$

(ii) $$\int \frac{1}{x^{10}} dx = \int x^{-10} dx = \frac{x^{-10+1}}{-10+1} + c = -\frac{1}{9x^9} + c$$

(iii) $$\int \sqrt{x} dx = \int x^{\frac{1}{2}} dx = \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} + c = \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + c = \frac{2}{3} x^{\frac{3}{2}} + c$$

(iv) $$\int \frac{1}{\sqrt{x}} dx = \int x^{-\frac{1}{2}} dx = \frac{x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} = 2\sqrt{x} + c$$